Appendix B: Feature Selection

Review note: this appendix replaces the 'work in progress' placeholder; all of its content is new.

This appendix documents the mutual-information (MI) feature selection described in Section 3.6 and tests how sensitive its outcome is to the main estimation choices. All results are recomputed from the candidate feature file with the procedure implemented in the feature-selection notebook. The recomputation recovers all 32 representative features used in the models.

B.1: Screening of the candidate features

Table B.1 traces the candidate features from the initial universe to the final predictor set. The missing-value screen is applied before the clustering; Table B.2 lists the 15 features it removes. The MI distances are estimated on the 680 weekly observations from 26 March 2012 to 31 March 2025, the period in which every remaining feature is available after forward filling.

Table B.1: From candidate features to the final predictor set

Step	Features
Candidate features in the feature file	359
Removed: more than 30% of weekly observations missing (Table B.2)	−15
Passing the missing-value screen	344
Entering the MI distance matrix (LocalGlobalPEFwdSpreadQS omitted by the implementation; Section 3.6)	343
Clusters at a cut height of 2, one representative each (Table B.3)	32
Relative-factor features added by design (Section 3.6)	18
Final predictor set	50

Table B.2: Features removed by the missing-value screen

Feature	Share of weeks missing	First available
SAImplYld_QS_Level	64%	05 Nov 2018
SAImplYld_QS_Slope	64%	05 Nov 2018
USDZAR_Vol_QS_Level	48%	30 Nov 2015
USDZAR_Vol_QS_Slope	48%	30 Nov 2015
SAImplYld_Level	43%	12 Jan 2015
SAImplYld_Slope	43%	12 Jan 2015
ZARI2M.Curncy_QS	33%	04 Feb 2013
ZARI2W.Curncy_QS	33%	04 Feb 2013
ZARI2Y.Curncy_QS	33%	04 Feb 2013
ZARI3M.Curncy_QS	33%	04 Feb 2013
ZARI3Y.Curncy_QS	33%	04 Feb 2013
ZARI4M.Curncy_QS	33%	04 Feb 2013
ZARI4Y.Curncy_QS	33%	04 Feb 2013
ZARI5M.Curncy_QS	33%	04 Feb 2013
ZARI5Y.Curncy_QS	33%	04 Feb 2013

The screen does not remove features without variation. Ten of the clustered features are constant over the clustering sample: the Nelson–Siegel decay parameters (Lambda) of the South African and US nominal and real yield curves and of the South African CDS curve, in level and quantile-scaled form (SA_NB_Lambda, SA_NB_Lambda_QS, SA_RB_Lambda, SA_RB_Lambda_QS, US_NB_Lambda, US_NB_Lambda_QS, US_RB_Lambda, US_RB_Lambda_QS, SA_CDS_Lambda, SA_CDS_Lambda_QS). Their entropy is zero, so their normalised MI with every other feature is zero and they lie at the maximum distance of one from all other features. They carry no information and should be removed before estimation; Section B.2 shows how they affect one selected representative.

B.2: Cluster membership and representatives

Table B.3 lists the 32 clusters formed at a cut height of 2, ordered by size. For each cluster it reports the representative retained for modelling, its category in Table 1, the cluster size, the representative's mean MI distance to the members of its cluster (Equation 7), the mean distance between all pairs of members, and the remaining members in order of increasing mean distance. Clusters with low within-cluster distances group closely related series, such as the same instrument at different tenors (ZAR forward points, USDZAR implied volatility and the VIX term structure). Seven two-member clusters have a distance of zero because they pair duplicated series, such as gold ETF statistics (GLD_*) and the corresponding gold spot statistics (XAU*). The largest clusters have within-cluster distances close to the median of 0.85 across all feature pairs: their members are only weakly related, so their representatives summarise a heterogeneous group rather than a set of near-duplicates.

One selected representative is itself nearly uninformative. PPP.ZA.Index_QS takes only the values 0.48, 0.74, 1.00; it equals 1.00 in every week until 30 January 2023 and varies only within the evaluation block. It is selected because it is the only member of its cluster with any variation: the other ten members are the constant Lambda features (within-cluster distance 1.00). It therefore carries no information in the initial training window and does not appear among the 20 most influential features of any model in Chapter 4.

Table B.3: Clusters, representatives and members (cut height 2)

Representative	Category	Size	Rep. mean dist.	Within-cluster dist.	Other members
GlobalBondVol6	Price-based features	48	0.77	0.81	US_RB_Level_TS; US_NB_Slope_TS; SubIndex_PC2; BCMPUSGR.Index_QS; SA_RB_Slope_TS; SABreakEvenInflPCA_TS; SA_NB_Level_TS; GlobalPropertyVol3; AAII_Bull_TS; LocalPropertyGret12; USDZARVol6; GlobalEquityVol3; US_NB_Level_TS; WOG1_Ent_TS; ACWI_Ent_TS; LocalPropertyVol3; InflGrowthSurpPC1_QS; InflGrowthSurpPC2; VolSpread6M; InflGrowthSurpPC2_QS; OVXStructBreak; JALSH_Ent_TS; BCMPUSIF.Index_QS; US_NB_Curvature_QS; SABreakEvenInflPCA_TS.1; US_NB_R2; DXY_RV_3; US_NB_Slope_QS; RepSouCDSStructBreak; LocalPropertyTS; SA_NB_Slope_TS; GlobalBondVol3; GlobalPropertyGret12; LocalEquityVol3; GlobalBondGret12; LocalBondsVol3; JALSH.Index.Forward.PE_QS; US_NB_R2_QS; GlobalPropertyTS; AAIIBULL.Index; NG1.Comdty_Diff_QS; SubIndex_PC1_TS; AAIIBEAR.Index; LocalPropertyGret6; JALSH_RV_3_TS; VIXStructBreak; USDZARVol3
VolTermStructPC2	Price-based features	44	0.86	0.88	GenComIndexPC2_QS; GenComIndexPC2; GlobalEquityGret1; XAU.Curncy_Diff_QS; GlobalPropertyVol1; VolTermStructPC2_QS; LA1RollSkew; USDZARGret3; USNomBndStructBreak_Level; LA1RollKurt; LocalPropertyVol1; LocalBondsGret3; GlobalPropertyGret1; TINSALNL.Index; LocalBondsVol1; GlobalBondVol1; LocalPropertyGret1; GlobalEquitySB; GlobalBondSB.1; SANomBndStructBreak_Level; TINSALNL.Index_QS; LocalEquityVol1; AAIIBULL.Index_QS; OPCVTPCR.Index; USRealBndStructBreak_Level; USDZARSB; LocalBondsGret1; SA_PMI_TS; OPCVTPCR.Index_QS; LocalBondsSB; MOVEStructBreak; LocalPropertySB; LocalEquityGret1; GlobalBondSB; USNomBndStructBreak_Slope; AAIIBEAR.Index_QS; GlobalBondGret1; USDZARVol1; USDZARGret1; SANomBndStructBreak_Slope; LocalEquitySB; USRealBndStructBreak_Slope; LA1StructBreak
JALSH_Fundamental_PC1	Fundamental	42	0.69	0.75	JALSH.Index.PE.Ratio_Diff; GE_GB_Corr; US_NB_Level; JALSH_PEvAgrgtPE_1FY; GlobalPropertyVol6; JALSH_RV_12_TS; InflGrowthSurpPC1; LocalPropertyVol6; SubIndex_PC1; SA_NB_Curvature; GlobalEquityVol6; SA_RB_Level_TS; JALSH.Index.Forward.PE_Diff; DXY_RV_12; MXWD_PEvAgrgtPE_1FY; ZAR_FwdPts_PC2_TS; US_RB_Level; LocalEquityVol6; BCMPEBLS.Index_QS; JALSH.Index.PE.Ratio_QS; ACWI_RV_12_TS; MXWD_Fundamental_PC2; MXWD_PEvFwdPE; TOP40.Index.PE.Ratio_Diff; LE_LB_Corr; SABreakEvenInflPC_1; ZAR_FwdPts_PC1_TS; LocalBondsVol6; LB_GB_Corr; JALSH_Fundamental_PC2; DXY_TS; LEI.Totl_TS; JALSH_PEvFwdPE; US_NB_Level_QS; US_RB_Slope_TS; SAFwdPts_Slope; US_RB_Level_QS; OEZAKLAF.Index_QS; SA_NB_Curvature_QS; TOP40.Index.PE.Ratio_QS; SA_RB_Level_QS
GlobalEquityTS	Sentiment Indicators	39	0.81	0.85	LocalEquityTS; GlobalBondTS; DXY.Curncy_QS; GlobalEquityGret12; GlobalEquityGret6; GlobalBondGret6; BICLB10Y.Index_QS; GlobalEquityGret3; LocalBondsTS; SA_NB_Level_QS; GE_GB_Corr.1; ACWI_RV_3_TS; LocalEquityGret12; LA1RollSD; GlobalBondGret3; MichgnConcIndx_QuantileScale; SABreakEvenInflPC_2; US_RB_R2; GlobalPropertyGret6; ECSURPUS.Index_QS; GE_LE_Corr; LocalEquityGret6; LocalBondsGret12; DXY_QuantScale; LocalGlobalPESpreadQS; GlobalPropertyGret3; LocalPropertyGret3; GB_LB_Corr; LocalEquityGret3; USDEUR_Vol_Slope; LocalBondsGret6; BASPCAAA.Index_QS; LE_LB_Corr.1; SAFwdPts_QS_Slope; US_RB_R2_QS; VolTermStructPC1_QS; USDEUR_Vol_QS_Slope; MichgnConcIndx_TS
SA_NB_Slope	Macro-economic	16	0.74	0.81	SA_CDS_Slope_QS; RandPPP_Factor_Inst; SA_CDS_Curvature_QS; USDZARV4Y.Curncy_QS; RandPPP_QuantileScale; SA_CDS_R2_QS; SA_CDS_Level_QS; USDZARV3Y.Curncy_QS; SA_NB_Slope_QS; SA_NB_R2; SA_NB_R2_QS; USDZARGret12; USDZARTS; USDZARGret6; USDZAR_Vol_Level
JALSH.Index.Price.to.Sales.Ratio_Diff	Fundamental	14	0.71	0.80	JALSH.Index.PE.Ratio.Next.FY.Aggregated_Diff; TOP40.Index.Forward.PE_Diff; TOP40.Index.PE.Ratio.Next.FY.Aggregated_Diff; TOP40.Index.Price.to.Sales.Ratio_Diff; JALSH.Index.PE.Ratio.Next.FY.Aggregated_QS; JALSH.Index.Price.to.Sales.Ratio_QS; TOP40.Index.Price.to.Sales.Ratio_QS; TOP40.Index.PE.Ratio.Next.FY.Aggregated_QS; TOP40.Index.Forward.PE_QS; VixMoveSpread; USDZAR_Vol_Slope; SABreakEvenInflPC_3; LEI.TOTL.Index_QS
GlobalPropertyVol12	Price-based features	12	0.58	0.66	GlobalEquityVol12; LocalPropertyVol12; LocalEquityVol12; GlobalBondVol12; LocalBondsVol12; SA_RB_Level; GlobalEquityVol1; US_NB_Curvature; SAFwdPtsl_Level; US_NB_Slope; USDZARVol12

Representative	Category	Size	Rep. mean dist.	Within-cluster dist.	Other members
PPP.ZA.Index_QS	Macro-economic	11	0.91	1.00	SA_NB_Lambda; SA_NB_Lambda_QS; SA_RB_Lambda; SA_RB_Lambda_QS; US_NB_Lambda; US_NB_Lambda_QS; US_RB_Lambda; US_RB_Lambda_QS; SA_CDS_Lambda; SA_CDS_Lambda_QS
GlobalEquity_EntDyn	Sentiment Indicators	10	0.82	0.93	GlobalEquity_Ent; GlobalBonds_Ent; LocalEquity_EntDyn; LocalEquity_Ent; LA1LempZiev; LocalBonds_Ent; MultiAssetEnt; USDZAR_Ent; GlobalBonds_EntDyn
MichgnConcIndx	Sentiment Indicators	10	0.56	0.65	CONCCONF.Index; EPUCCUSM.Index; CONSSENT.Index_QS; EPUCCUSM.Index_QS; SAPMI_YoY; CONCCONF.Index_QS; SACWC.Index_QS; SAPMI.Index_QS; SSICCONF.Index_QS
EURUSDV6M.Curncy_QS	Price-based features	10	0.50	0.63	EURUSDV9M.Curncy_QS; EURUSDV3M.Curncy_QS; EURUSDV2M.Curncy_QS; EURUSDV1Y.Curncy_QS; EURUSDV1M.Curncy_QS; EURUSDV2W.Curncy_QS; USDEUR_Vol_QS_Level; EURUSDV1W.Curncy_QS; USDEUR_Vol_Level
ZAR18M.Curncy_QS	Macro-economic	9	0.23	0.30	ZAR2Y.Curncy_QS; ZAR12M.Curncy_QS; SAFwdPts_QS_Level; ZAR10M.Curncy_QS; ZAR15M.Curncy_QS; ZAR3Y.Curncy_QS; ZAR21M.Curncy_QS; ZAR5Y.Curncy_QS
MXWD.Index.PE.Ratio_QS	Fundamental	8	0.66	0.76	MXWD.Index.Price.to.Ebitda_QS; MXWD.Index.Price.to.Sales.Ratio_Diff; MXWD.Index.Price.to.Book.Ratio_Diff; MXWD.Index.Price.to.Sales.Ratio_QS; MXWD.Index.Forward.PE_QS; MXWD.Index.PE.Ratio.Next.FY.Aggregated_QS; MXWD.Index.Price.to.Book.Ratio_QS
USDZARV1M.Curncy_QS	Price-based features	6	0.46	0.60	USDZARV2M.Curncy_QS; USDZARV3W.Curncy_QS; USDZARV3M.Curncy_QS; USDZARV1W.Curncy_QS; USDZARVON.Curncy_QS
SA_RB_Curvature	Macro-economic	6	0.50	0.64	SA_RB_Slope; SA_RB_Slope_QS; SA_RB_Curvature_QS; SA_RB_R2; SA_RB_R2_QS
GenComIndexPC1_QS	Sentiment Indicators	6	0.55	0.75	LA1.Comdty_Diff_QS; GenComIndexPC1; LA1FracDiff; HG1.Comdty_Diff_QS; XAG.Curncy_Diff_QS
VolTermStructPC1	Price-based features	5	0.35	0.50	VIX3M.Index; VIX6M.Index; VIX1Y.Index; VIX.Index
USDZARV1Y.Curncy_QS	Price-based features	5	0.31	0.42	USDZARV18M.Curncy_QS; USDZARV9M.Curncy_QS; USDZARV2Y.Curncy_QS; USDZARV6M.Curncy_QS
SA_CDS_Slope	Macro-economic	5	0.45	0.61	SA_CDS_Curvature; SA_CDS_Level; SA_CDS_R2; SA_NB_Level
MXWD_Fundamental_PC1	Fundamental	5	0.41	0.57	MXWD.Index.Forward.PE_Diff; MXWD.Index.PE.Ratio_Diff; MXWD.Index.PE.Ratio.Next.FY.Aggregated_Diff; MXWD.Index.Price.to.Ebitda_Diff
TOP40.Index.Price.to.Ebitda_Diff	Fundamental	4	0.43	0.58	JALSH.Index.Price.to.Ebitda_Diff; TOP40.Index.Price.to.Ebitda_QS; JALSH.Index.Price.to.Ebitda_QS
BCMPEBLS.Index	Macro-economic	4	0.32	0.47	SACWC_YoY; LEI.TOTL.Index; OEZAKLAF_YoY
US_RB_Curvature	Macro-economic	4	0.34	0.46	US_RB_Slope; US_RB_Slope_QS; US_RB_Curvature_QS
TOP40.Index.Price.to.Book.Ratio_Diff	Fundamental	4	0.51	0.69	JALSH.Index.Price.to.Book.Ratio_Diff; TOP40.Index.Price.to.Book.Ratio_QS; JALSH.Index.Price.to.Book.Ratio_QS
GLD_RollKurt	Sentiment Indicators	2	0.00	0.00	XAURollKurt
GLD_LempZiev	Sentiment Indicators	2	0.06	0.11	XAULempZiev
GLD_RollSD	Price-based features	2	0.00	0.00	XAURollSD
GLD_StructBreak	Sentiment Indicators	2	0.00	0.00	XAUStructBreak
USDZAR_EntDyn	Macro-economic	2	0.00	0.00	LocalBonds_EntDyn
GLD_FracDiff	Price-based features	2	0.00	0.00	XAUFracDiff
RandPPP_Factor_Inst_TS_ST	Macro-economic	2	0.00	0.00	RandPPP_Factor_Inst_TS_LT
GLD_RollSkew	Sentiment Indicators	2	0.00	0.00	XAURollSkew

B.3: Choice of cut height

Table B.4 reports the number of clusters and the mean silhouette score for a range of cut heights, and Figure B.1 plots the silhouette score for partitions of 5 to 100 clusters. The silhouette score compares each feature's mean distance to the other members of its cluster with its mean distance to the nearest other cluster; values near zero indicate overlapping clusters. The scores remain below 0.1 throughout, at 0.037 for the 32 clusters used. Beyond about 20 clusters they increase with the number of clusters, reaching 0.097 at 100 clusters, the largest partition considered, without an interior maximum. The data therefore do not suggest a natural number of clusters, and the cut height of 2 reflects a judgement that balances a compact predictor set against merging dissimilar features.

Table B.4: Number of clusters and silhouette score by cut height

Cut height	Clusters	Mean silhouette score
1	265	0.115
1.5	65	0.077
2	32	0.037
2.5	18	0.022
3	11	-0.011
4	8	0.023

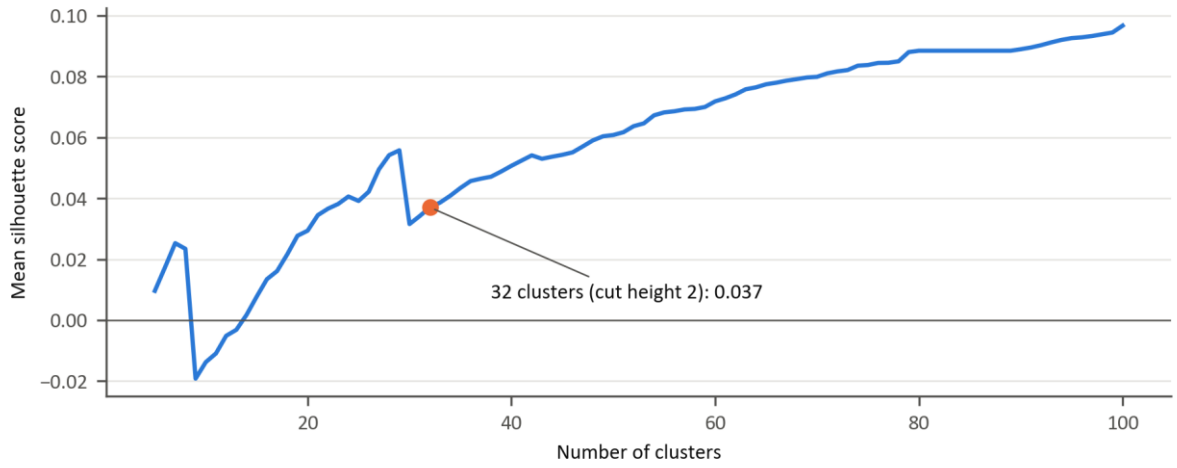


Figure B.1: Mean silhouette score by number of clusters (MI distances, Ward linkage)

B.4: Stability of the selected features

Table B.5 repeats the selection under four alternative choices: correcting the omission of the final feature column; 10 and 30 quantile bins instead of 20; and estimating the MI distances on the 514 weeks before the evaluation block only (to 24 January 2022), which removes evaluation-period information from the selection. Changing the number of bins or the sample also changes the scale of the linkage heights, so each alternative is compared with the baseline at the same number of clusters (32). A

representative is counted as substituted when the alternative selection replaces it with another member of its baseline cluster. The adjusted Rand index (ARI) measures agreement between the baseline and alternative cluster memberships (one for identical partitions, zero for chance agreement).

Table B.5: Representatives retained under alternative estimation choices (32 clusters)

Alternative	Clusters at cut height 2	Unchanged	Substituted from same cluster	Not represented	ARI
All 344 screened features (omission corrected)	32	31	0	1	0.91
10 quantile bins instead of 20	40	20	8	4	0.39
30 quantile bins instead of 20	23	23	7	2	0.59
Pre-evaluation weeks only (Mar 2012 to Jan 2022)	31	21	9	2	0.53

Correcting the omitted column changes one representative: VolTermStructPC2 is replaced by GlobalEquityGret3, a near tie in the largest cluster. The other choices change between 9 and 12 of the 32 representatives. Most replacements come from the same baseline cluster, so between 28 and 30 of the 32 baseline clusters remain represented. The specific features selected are therefore sensitive to reasonable estimation choices, while the groups of information they represent are largely stable. Restricting the estimation to pre-evaluation data changes 11 representatives, 9 of which are replaced from the same cluster. The full-sample estimation disclosed in Section 3.6 therefore influenced which specific features were selected. Whether this affects out-of-sample performance could only be established by re-estimating the models with the pre-evaluation selection. Table B.6 lists the changes. With 10 bins, one cluster of constant Lambda features is represented by one of them (SA_NB_Lambda), which illustrates why constant features should be removed before clustering.

Table B.6: Representatives removed and added under each alternative (32 clusters)

Alternative	Removed from the selection	Added to the selection
All 344 screened features (omission corrected)	VolTermStructPC2	GlobalEquityGret3
10 quantile bins instead of 20	BCMPEBLS.Index; GlobalBondVol6; GlobalEquityTS; GlobalEquity_EntDyn; MXWD.Index.PE.Ratio_QS; MichgnConcIndx; PPP.ZA.Index_QS; SA_CDS_Slope; SA_RB_Curvature; USDZARV1Y.Curncy_QS; VolTermStructPC2; ZAR18M.Curncy_QS	GlobalEquityGret1; GlobalEquityVol3; LocalEquityTS; MXWD.Index.Price.to.Ebitda_QS; MXWD_PEvAgrgtPE_1FY; SA_NB_Lambda; SA_RB_Level; SA_RB_Slope; USDZARV18M.Curncy_QS; US_NB_Level; ZAR2Y.Curncy_QS; ZAR_FwdPts_PC1_TS
30 quantile bins instead of 20	GlobalEquityTS; GlobalPropertyVol12; JALSH.Index.Price.to.Sales.Ratio_Diff; MXWD.Index.PE.Ratio_QS; MichgnConcIndx; SA_RB_Curvature; TOP40.Index.Price.to.Book.Ratio_Diff; TOP40.Index.Price.to.Ebitda_Diff; VolTermStructPC2	CONCCONF.Index; GenComIndexPC2; GlobalEquityVol12; JALSH.Index.Price.to.Book.Ratio_Diff; JALSH.Index.Price.to.Ebitda_Diff; LEI.TOTL.Index_QS; MXWD.Index.Price.to.Sales.Ratio_Diff; OEZAKLAF.Index_QS; USDEUR_Vol_QS_Level
Pre-evaluation weeks only (Mar 2012 to Jan 2022)	GlobalBondVol6; GlobalEquityTS; GlobalEquity_EntDyn; GlobalPropertyVol12; JALSH.Index.Price.to.Sales.Ratio_Diff; JALSH_Fundamental_PC1; MXWD.Index.PE.Ratio_QS; MichgnConcIndx; TOP40.Index.Price.to.Ebitda_Diff; US_RB_Curvature; VolTermStructPC2	CONCCONF.Index; GenComIndexPC2_QS; GlobalBondTS; GlobalBonds_EntDyn; GlobalPropertyVol1; JALSH.Index.Price.to.Ebitda_Diff; MXWD.Index.Price.to.Sales.Ratio_Diff; USDZARV4Y.Curncy_QS; US_NB_Curvature; US_NB_Level; US_RB_Slope

B.5: Mutual information compared with correlation

Section 3.6 motivates MI by its ability to capture nonlinear dependence that Pearson correlation misses. To test whether the choice of dependence measure matters, the clustering is repeated with the distance $1 - |r|$, using the same sample, Ward linkage and number of clusters. Only 7 of the 32 MI representatives are also selected under correlation, a further 14 are replaced by a member of the same MI cluster, and the ARI is 0.15. The choice of dependence measure therefore changes the selection materially.

Figure B.2 compares the two measures across all 58,653 feature pairs. They are related (Spearman rank correlation 0.46) but diverge in two systematic ways. First, the 14 pairs with high MI (NMI ≥ 0.4) and low correlation ($|r| < 0.2$) all involve slow-moving series, with a median week-to-week autocorrelation of 0.97. When MI is recomputed on weekly changes, their median NMI falls from 0.42 to 0.04 (Table B.7). Their high MI reflects series that occupy similar ranges during the same periods, rather than a stable nonlinear relationship. Second, the 287 strongly correlated pairs ($|r| \geq 0.8$) have a median NMI of 0.38, and 74 of them fall below 0.3: with 20 bins, normalising by entropy places strong linear dependence well below the top of the NMI scale, so pairs that correlation treats as largely redundant can remain far apart in MI distance.

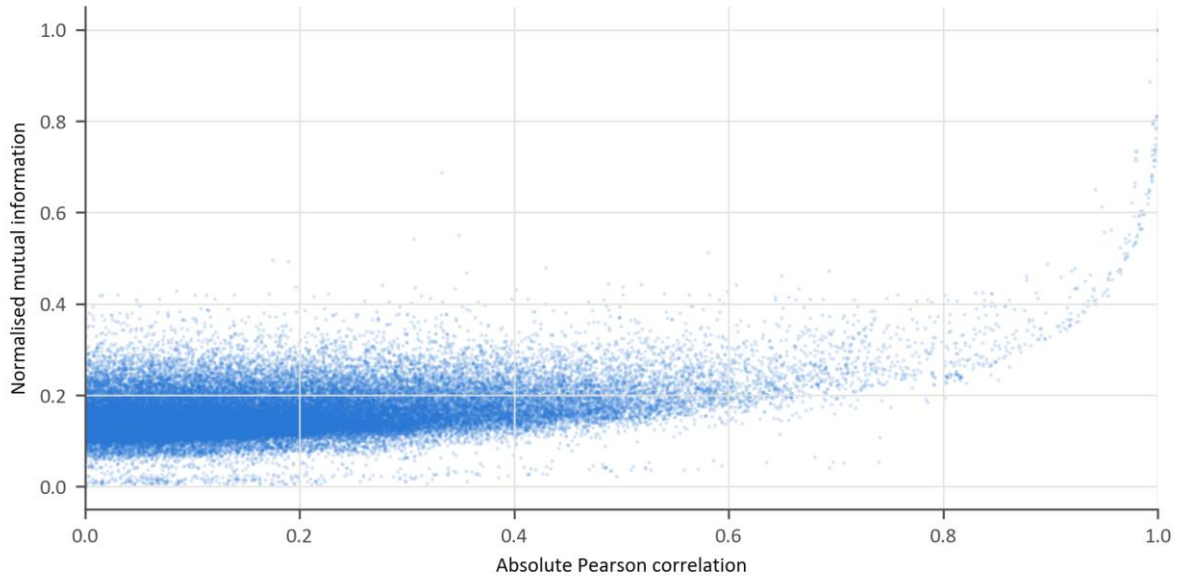


Figure B.2: Normalised MI against absolute Pearson correlation for all 58,653 pairs of clustered features

Table B.7: Feature pairs with high MI but low correlation

Feature pair	NMI (levels)	NMI (weekly changes)	|r| (levels)	Lag-1 autocorrelation
BCMPEBLS.Index – SACWC.Index_QS	0.50	0.14	0.17	0.98 / 0.97
OEZAKLAF_YoY – BCMPEBLS.Index	0.49	0.09	0.19	0.98 / 0.98
BCMPEBLS.Index – LocalPropertyVol12	0.44	0.01	0.20	0.98 / 0.99
LEI.TOTL.Index – SACWC.Index_QS	0.43	0.06	0.09	0.99 / 0.97
SACWC_YoY – JALSH_Fundamental_PC1	0.42	0.01	0.17	0.97 / 0.99
SACWC_YoY – USDZARVol12	0.42	0.00	0.03	0.97 / 0.99
BCMPEBLS.Index – CONCCONF.Index	0.42	0.11	0.14	0.98 / 0.96
BCMPEBLS.Index – JALSH_PEvAgrgtPE_1FY	0.42	0.03	0.02	0.98 / 0.97
BCMPEBLS.Index – SA_RB_Level	0.42	0.01	0.10	0.98 / 0.99
BCMPEBLS.Index – JALSH_Fundamental_PC1	0.42	0.01	0.11	0.98 / 0.99
SACWC_YoY – SA_RB_Level	0.42	0.00	0.01	0.97 / 0.99
MichgnConcIndx – BCMPEBLS.Index	0.41	0.10	0.07	0.99 / 0.98
MichgnConcIndx – SACWC_YoY	0.41	0.07	0.12	0.99 / 0.97
MichgnConcIndx – OEZAKLAF_YoY	0.41	0.32	0.05	0.99 / 0.98

MI-based and correlation-based clustering therefore produce materially different selections in this dataset, but the evidence does not show that the difference arises from nonlinear redundancy. Much of the dependence that MI detects and correlation does not is shared persistence between slow-moving series.

B.6: Summary

The documented procedure reproduces the selected features exactly. The selection is sensitive to the bin count, the estimation sample and the dependence measure: about a third of the representatives

change under each reasonable alternative, although most are replaced by features from the same cluster. The clustering is weakly structured, and part of the MI it relies on reflects shared persistence rather than nonlinear dependence. The candidate universe also contains ten constant features, which leave one nearly uninformative representative (PPP.ZA.Index_QS) in the predictor set. The selected features should therefore be read as one reasonable, non-redundant subset of the candidate universe, not as a uniquely identified set of informative predictors.